2026

TdP

Python and Databases

Access, DAO Pattern, ORM, Identity Map, Pooling

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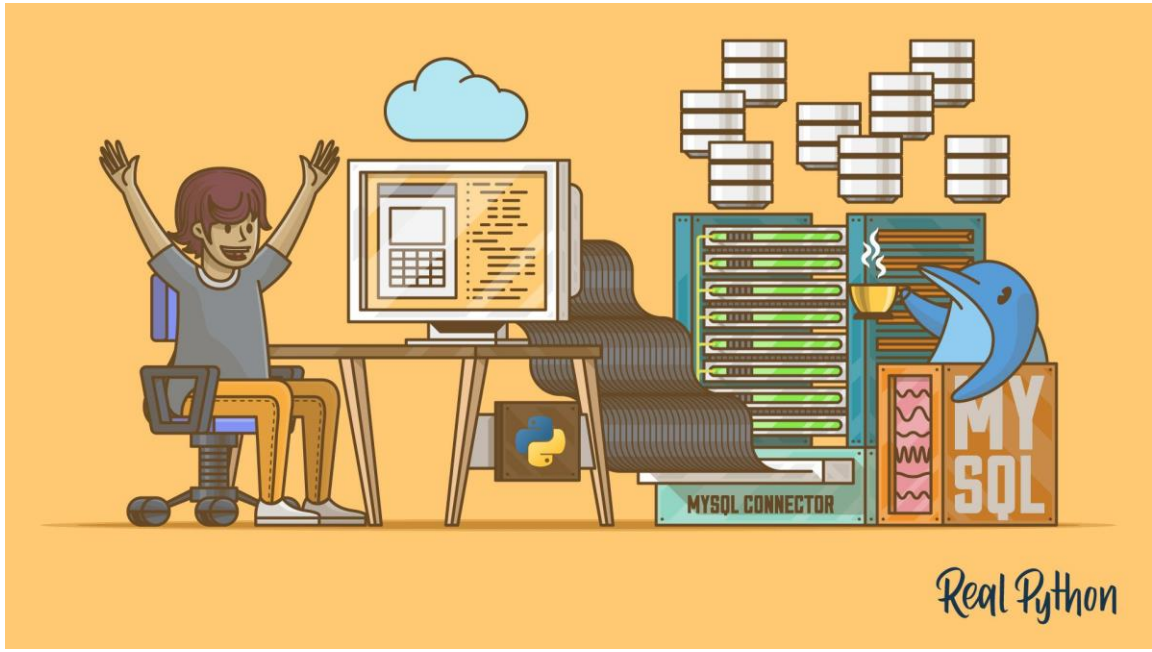

https://realpython.com/python-mysql/
https://realpython.com/python-sql-libraries/
https://realpython.com/python-pyqt-database/

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Outline

- Tools
  - MariaDB, MySQL, DBeaver
- Database access in Python
  - mysql-connector, Connection, Cursor, Statements
- Pattern DAO
- Object-Relational Mapping (ORM)
- Connection Pooling

Goal

- Enable Python applications to access data stored in Relational Databases
  - Query existing data
  - Modify existing data
  - Insert new data
- The data can be used by
  - The algorithms running in the application
  - The user, through the user interface

TOOLS

http://dilbert.com/strips/comic/1995-11-17/

Tools: MariaDB / MySQL

Database Management Systems (DBMS) are software systems used to store, retrieve, and run queries on data, as well as administer the data. A DBMS allows end-users/applications to interact with a database.

MariaDB Foundation

https://mariadb.org/

MySQL®

https://www.mysql.com/

Tools: DBeaver

Graphical frontend to work with a database:

- Data Editor
- SQL Editor
- Task management
- Database maintenance tools
- NB: STICK WITH VERSION SUGGESTED
  - Dbeaver 23.x

https://dbeaver.io/

Tools: MySQL Workbench

Graphical frontend to work with a database:

- Data Editor
- SQL Editor
- Task management
- Database maintenance tools
- Slightly better supported

https://dev.mysql.com/downloads/workbench/

mySQL-connector

DATABASE ACCESS IN PYTHON

Resources

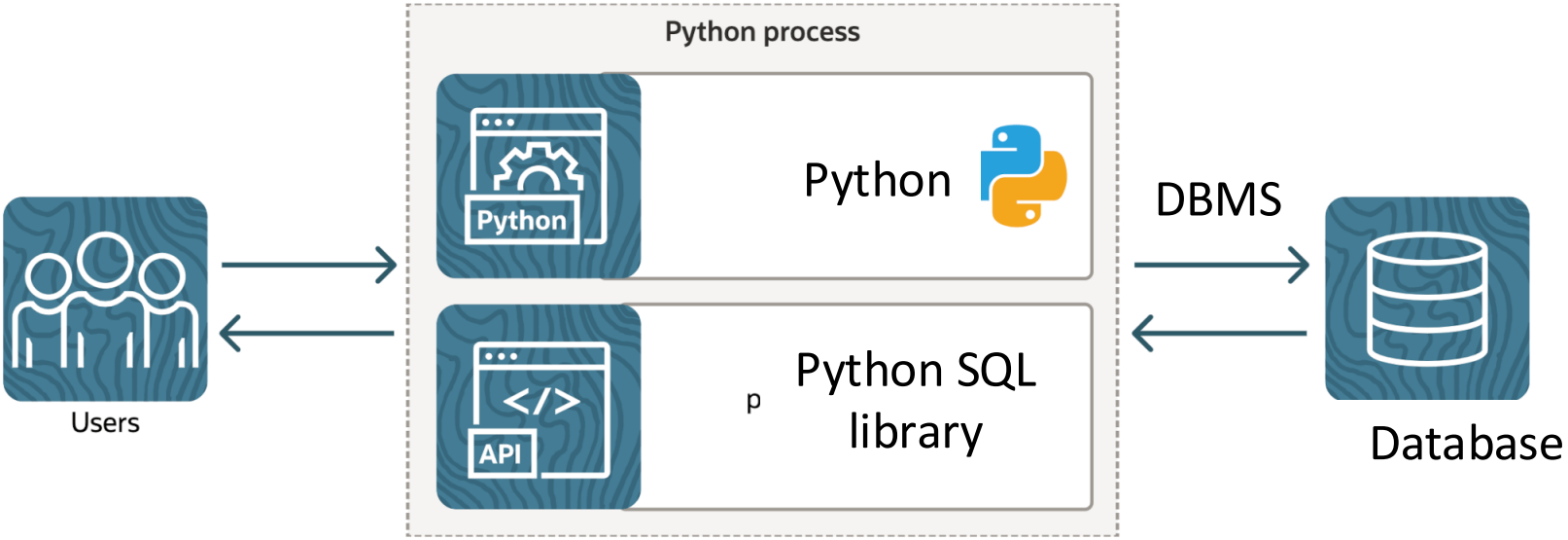
- Official **mySQL-connector** guide: https://dev.mysql.com/doc/connector-python/en/
- Useful tutorials: https://www.geeksforgeeks.org/how-to-connect-python-with-sql-database/

Connecting and interacting with DBMS

- **Database management system** (DBMS): software system that enables users to define, create, maintain and control access to the database
- Different flavours of SQL-based DBMSs: MySQL, MariaDB, PostgreSQL, SQLite, and SQL Server, …
  - All of these databases are compliant with the SQL standards but with varying degrees of compliance

https://troels.arvin.dk/db/rdbms/

Interacting with DBMS in Python



Many different Python libraries (**connectors**) that implement modules for interacting with different DBMS

- `mysql-connector-python`
- `SQLite`
- `Psycopg2`
- `mariadb-connector-python`
- …

https://realpython.com/python-sql-libraries

Python Database API Specification

The Python Database API (DB-API) defines a standard interface for Python database access modules, e.g., using **Connection** and **Cursor** objects.

Goals:

- Encourage similarity between the Python modules that are used to access databases.
- Achieve a consistency leading to more easily understood modules.
- Code that is generally more portable across databases.

https://peps.python.org/pep-0249/

mysql-connector-python

- It is a self-contained Python driver for communicating with MySQL servers, using an API that is compliant with the Python Database API Specification v2.0 (PEP 249)
- Documentation: https://dev.mysql.com/doc/connector-python/en/
- Install via pip (`pip install mysql-connector-python`) , or directly in PyCharm in the virtual environment of the project

Connection

- The first step in interfacing a Python application with a MySQL/MariaDB server is to establish a connection
- mysql-connector provides a `connect()` function that is used to establish connections to the MySQL server

```
import mysql.connector

cnx = mysql.connector.connect(user='scott',
                              password='password',
                              host='127.0.0.1',
                              database='employees')
cnx.close()
```

https://dev.mysql.com/doc/connector-python/en/connector-python-example-connecting.html

Connection

```
import mysql.connector

cnx = mysql.connector.connect(user='scott',
                              password='password',
                              host='127.0.0.1',
                              database='employees')

cnx.close()
```

It is a `MySQLConnection` object.

The `MySQLConnection` class is used to open and manage a connection to a MySQL server. It also used to send commands and SQL statements and read the results.

The arguments of the `connect()` identify the database we want to connect, and the credentials of the user

There are many more arguments…

https://dev.mysql.com/doc/connector-python/en/connector-python-connectargs.html

Connection

```
import mysql.connector

cnx = mysql.connector.connect(user='scott',
                              password='password',
                              host='127.0.0.1',
                              database='employees')
cnx.close()
```

The `close()` function closes the connection, when we don't need it anymore.

- The connection to the database is a resource!!!

Connection

- The `connect()` function may raise exceptions (for example if the connection fails due to wrong authentication)

```
import mysql.connector
from mysql.connector import errorcode

try:
  cnx = mysql.connector.connect(user='scott',
                                database='employ')
except mysql.connector.Error as err:
  if err.errno == errorcode.ER_ACCESS_DENIED_ERROR:
    print("Something is wrong with your user name or password")
  elif err.errno == errorcode.ER_BAD_DB_ERROR:
    print("Database does not exist")
  else:
    print(err)
else:
  cnx.close()
```

We can handle these exceptions with a **try** – **except** – **else** - **finally** clause:

- 1. Try to connect
- 2. Handle exceptions
- 3. If there was no exception, close the connection

This may also be rewritten using a with statement

https://docs.python.org/3/reference/compound_stmts.html#with

Connection

- Writing the configuration of the database and authentication information in the code is not ideal, especially if the file is worked collaboratively (git)
- It is possible to use a separate config file.

```
cnx = mysql.connector.connect(option_files='/etc/mysql/connectors.cnf')
```

Example config file

```
[client]
user=John
password=Wick
host=127.0.0.1
database=tests
raise_on_warnings=True
```

Using the connection

- When connected to the DBMS and we have the `MySQLConnection` object, we can interact in different ways
  - Create tables
  - Create/Update/Delete data
  - Read data
- This is achieved through the execution of **SQL statements**, using a handle structure known as **cursor**

Tables creation: https://dev.mysql.com/doc/connector-python/en/connector-python-example-ddl.html

Cursor

Cursor default position (before first record)

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
102	Jyothi	7550.50	3rd Record

Cursor on first record

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
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Cursor position after last record

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
102	Jyothi	7550.50	3rd Record

Cursor

- The `MySQLCursor` class instantiates objects that can execute operations such as SQL statements. A cursor is created from a `MySQLConnection` using the `cursor()` function
- There are several cursor classes that inherit from the `MySQLCursor`, and can be created by passing an appropriate argument to the `cursor()` function

```
import mysql.connector

cnx = mysql.connector.connect(database='world')
cursor = cnx.cursor()
cursor_dict = cnx.cursor(dictionary=True)
cursor_tuple = cnx.cursor(named_tuple=True)
cursor_prepared = cnx.cursor(prepared=True)
```

`MySQLCursorDict` cursor returns rows as dictionaries

`MySQLCursorNamedTuple` cursor returns rows as named tuples

`MySQLCursorPrepared` cursor is used for executing prepared statements

Cursor documentation: https://dev.mysql.com/doc/connector-python/en/connector-python-api-mysqlcursor.html

Statement execution

- A cursor object has a method `execute()` that allows to execute a SQL statement, expressed as a string

```
query = """Select id, name from user"""
cursor.execute(query)
```

Parametric queries

- SQL queries may depend on user input data
- Example: find item whose code is specified by the user
- Method 1: string interpolation (with concatenation or as an f-string)
  -
```
query =
"SELECT * FROM items
WHERE code='"+user_code+”’” ;
```

Parametric queries

- SQL queries may depend on user input data
- Example: find item whose code is specified by the user
- Method 1: string interpolation (with concatenation or as an f-string)
  - query =
    "SELECT * FROM items
    WHERE code='"+user_code+”’” ;

Parametric queries

- SQL queries may depend on user input data
- Example: find item whose code is specified by the user
- Method 1: string interpolation (with concatenation or as an f-string)
  - `query = "SELECT * FROM items WHERE code='"+user_code+”’” ;`
- Method 2: use parametric Statements
  - Always preferable
  - Always

What’s wrong with method 1?

- ```
  query =
  "SELECT * FROM items
  WHERE code='"+user_code+"'" ;
  ```

For example, string written by the user in a textbox in the GUI

- This may cause security problems
  ```

SQL injection

- SQL injection – syntax errors or privilege escalation
- Example
  - username : `'; delete * from users ; --`

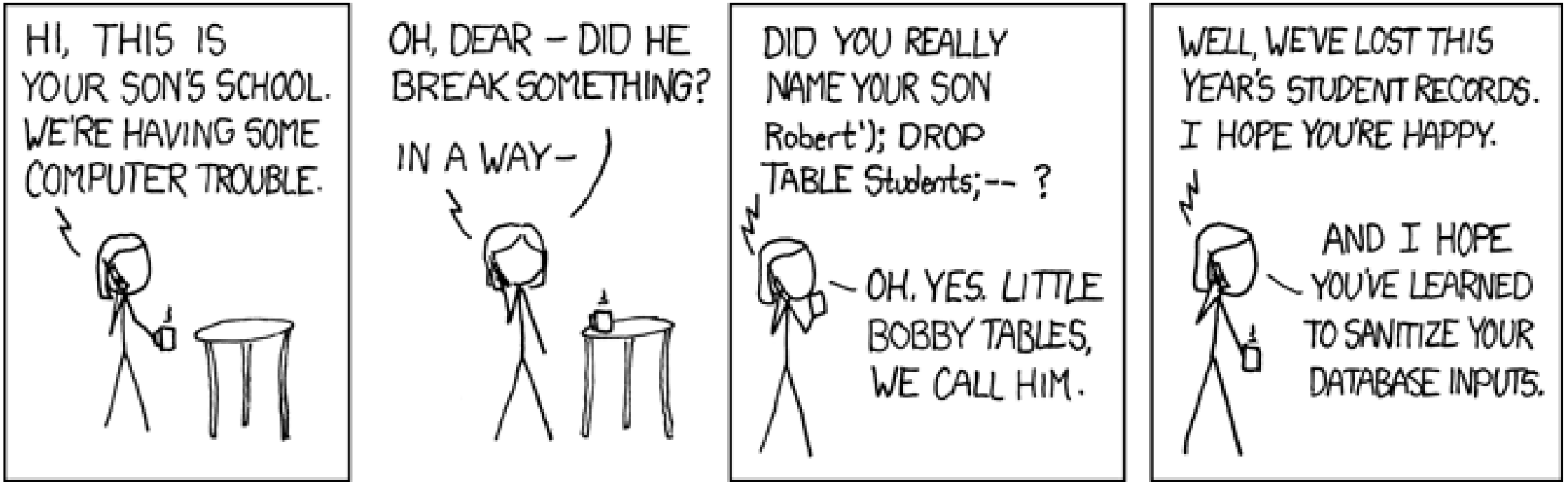
```
select * from users where
username=''; delete * from
users ; -- '
```

- **Must** detect or escape all dangerous characters!
  - Will **never** be perfect…

- **Never** trust user-entered data. Never. Not once. Really.

SQL injection attempt ☺

SQL injection attempt ☺



http://xkcd.com/327/

Parametric statements

- Separate statement **creation** from statement **execution**
  - At creation time: define SQL syntax (**template**), with placeholders for variable quantities (**parameters**)
  - At execution time: define actual quantities for placeholders (**parameter values**), and run the statement
- Parametric statements can be re-run many times
- Parameter values are automatically
  - Converted according to their primitive type
  - Escaped, if they contain dangerous characters
  - Handle non-character data (serialization)

Insert/Update/Delete data

- Using the cursor, we can execute INSERT, UPDATE and DELETE statements

```
import mysql.connector

cnx = mysql.connector.connect(user='John',
                              password='Wick',
                              host='127.0.0.1',
                              database='tests')
cursor = cnx.cursor()

add_test = """INSERT INTO tests
              (id, name, value)
              VALUES (%s, %s, %s)"""

cursor.execute(add_test, (1, "John Doe", 11.3))

cnx.commit()
cursor.close()
cnx.close()
```

- 1. Define the statement (using the Python multi-line block “”” “”””). Values may be written in the statement, or left unspecified as %s (because it may depend on user data)
- 2. Execute the statement (setting all the unspecified values)
- 3. **Commit the changes to the database**
- 4. Close the cursor

Insert/Update/Delete data

- Using the cursor, we can execute INSERT, UPDATE and DELETE statements

```
import mysql.connector

cnx = mysql.connector.connect(user='John',
                              password='Wick',
                              host='127.0.0.1',
                              database='tests')
cursor = cnx.cursor()

update_test = """UPDATE tests
                 SET value = %s
                 WHERE id = %s"""

cursor.execute(update_test, (99.9, 3))

cnx.commit()
cursor.close()
cnx.close()
```

```
import mysql.connector

cnx = mysql.connector.connect(user='John',
                              password='Wick',
                              host='127.0.0.1',
                              database='tests')
cursor = cnx.cursor()

delete_test = """DELETE FROM tests
                 WHERE id = %s"""

cursor.execute(delete_test, (5,))

cnx.commit()
cursor.close()
cnx.close()
```

Query Data

- We can use a cursor also to execute a statement that queries data from the database

```
import mysql.connector

cnx = mysql.connector.connect(user='John',
                              password='Wick',
                              host='127.0.0.1',
                              database='prova')
cursor = cnx.cursor()
query = """SELECT * FROM test"""
cursor.execute(query)

for (id, name, value) in cursor:
    print(id, name, value)

cursor.close()
```

- Executing the query fetches results from the database
- We can then use cursor as an iterator over the results set
- There is no commit, because we are not modifying the database

Query Data: process the results

- After executing the statement, we can use the cursor as a iterator to go through the results

Cursor default position (before first record)

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
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Cursor on first record

100	S N Rao	5500. 50	1st Record
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Cursor position after last record

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
102	Jyothi	7550.50	3rd Record

- Data is available a row at a time
- Rows are read as tuple, in the standard cursor. They can be read as dictionary or as named tuple using the corresponding cursor

Query Data

- We can use a cursor also to execute a statement that queries data from the database

```
import mysql.connector

cnx = mysql.connector.connect(user='John',
                              password='Wick',
                              host='127.0.0.1',
                              database='prova')
cursor = cnx.cursor(dictionary=True)
query = """SELECT * FROM test"""
cursor.execute(query)

for row in cursor:
    print(row["id"], row["name"], row["value"])

cursor.close()
```

- If we use a `MySQLCursorDict` the results set is read as a dictionary, so we can iterate through the data accordingly

Query Data: fetchone, fetchmany, fetchall

- The cursor object also has other methods to fetch the results retrived by exceuing a query statement
  - `fetchone()` retrieves the next row of a query result set and returns a single sequence, or `None` if no more rows are available
  - `fetchmany(N)` fetches the next set of N rows of a query result and returns a list of tuples (or dictionaries or named tuples, if using other specialized cursors)
  - `fetchall()` fetches all (or **all remaining**) rows of a query result set and returns a list of tuples (or dictionaries or named tuples, if using other specialized cursors). If no more rows are available, it returns an empty list.

Query Data: fetchone, fetchmany, fetchall

Cursor before the first row

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
102	Jyothi	7550.50	3rd Record

Cursor at the first row

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
102	Jyothi	7550.50	3rd Record

Cursor after the last row

100	S N Rao	5500. 50	1st Record
101	Jyostna	6500.50	2nd Record
102	Jyothi	7550.50	3rd Record

<u>Example</u>

- 0. cursor.execute(query)
- 1. cursor.fetchone()
- 2. cursor.fetchall()

Query Data: fetchone, fetchmany, fetchall

Example

```
import mysql.connector

cnx = mysql.connector.connect(user='John',
                              password='Wick',
                              host='127.0.0.1',
                              database='prova')
cursor = cnx.cursor(dictionary=True)
query = """SELECT * FROM test"""
cursor.execute(query)

rows = cursor.fetchall()
print(rows)

cursor.close()
```

Query Data: fetchone, fetchmany, fetchall

Warning: when executing a query to read the data, we are expected to handle all the results.

```python
query = """SELECT * from test"""
cursor.execute(query)

row1 = cursor.fetchone()
print(row1)

cursor.close()
cnx.close()
```

test_db

```
  File "/Users/carlo/TdP2024/test_db/pythonProject/test_db.py", line 49, in <module>
    cursor.close()
  File "/Users/carlo/TdP2024/test_db/pythonProject/.venv/lib/python3.12/site-packages/mys
    self._cnx.handle_unread_result()
  File "/Users/carlo/TdP2024/test_db/pythonProject/.venv/lib/python3.12/site-packages/mys
    raise InternalError("Unread result found")
mysql.connector.errors.InternalError: Unread result found
(1, 'prova carico', 99.9)
```

Query Data: fetchone, fetchmany, fetchall

```python
query = """SELECT * from test"
cursor.execute(query)

row = cursor.fetchone()
while row is not None:
    print(row)
    row = cursor.fetchone()

cursor.close()
```

while row is not None

Run test_db

```
/Users/carlo/TdP2024/test_db/pytho
(1, 'prova carico', 99.9)
(2, 'prova latenza', 1.0)
(3, 'test supporto', 3.4)
(4, 'test campo', 3.4)
```

```python
query = """SELECT * from test"""
cursor.execute(query)

row = cursor.fetchone()
print(row)
rows = cursor.fetchall()
print(rows)

cursor.close()
```

test_db

```
/Users/carlo/TdP2024/test_db/pythonProject/.venv/bin/python /Users/carlo/TdP202
(1, 'prova carico', 99.9)
[(2, 'prova latenza', 1.0), (3, 'test supporto', 3.4), (4, 'test campo', 3.4)]
```

Type conversion MySQL -> Python

- By default, MySQL types in result sets are converted automatically to Python types. For example, a `DATETIME` column value becomes a `datetime.datetime` object. To disable conversion, one can use a cursor with the option cursor(raw=True)
- You can check the read type using the Python `type()` function

DATA ACCESS OBJECT (DAO)

Problems

- Database code involves a lot of «specific» knowledge
  - Connection parameters
  - SQL commands
  - The structure of the database
- Bad practice to «mix» this low-level information with main application code
  - Reduces portability and maintainability
  - Creates more complex code
  - Breaks the «one-class one-task» assumption
- What is a better code organization?

Goals

- Encapsulate DataBase access into separate classes and modules, distinct from application ones
  - All other classes should be shielded from DB details
- DataBase access should be independent from application needs
  - Potentially reusable in different parts of the application
- Develop a reusable development pattern that can be easily applied to different situations

Pattern DAO

- DAO (Data Access Object) is a pattern that acts as an abstraction between the database and the main application.
- It takes care of adding, modifying, retrieving, and deleting the data and you do not need to know how it does this, that’s what an abstraction is.
- DAO is implemented in a separate file. Then, these methods are called in the main application.

Pattern DAO

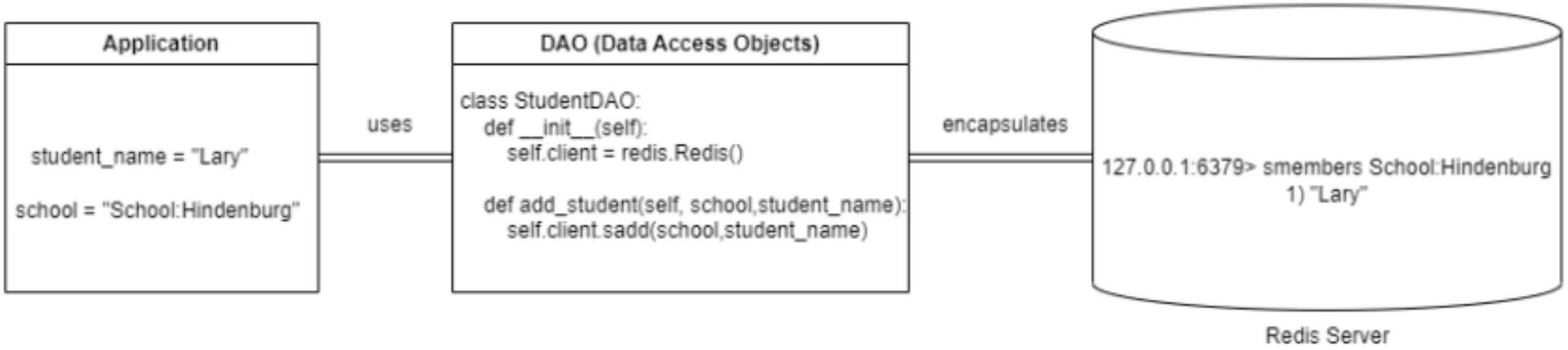


Image source https://www.analyticsvidhya.com/blog/2023/02/what-are-data-access-object-and-data-transfer-object-in-python/

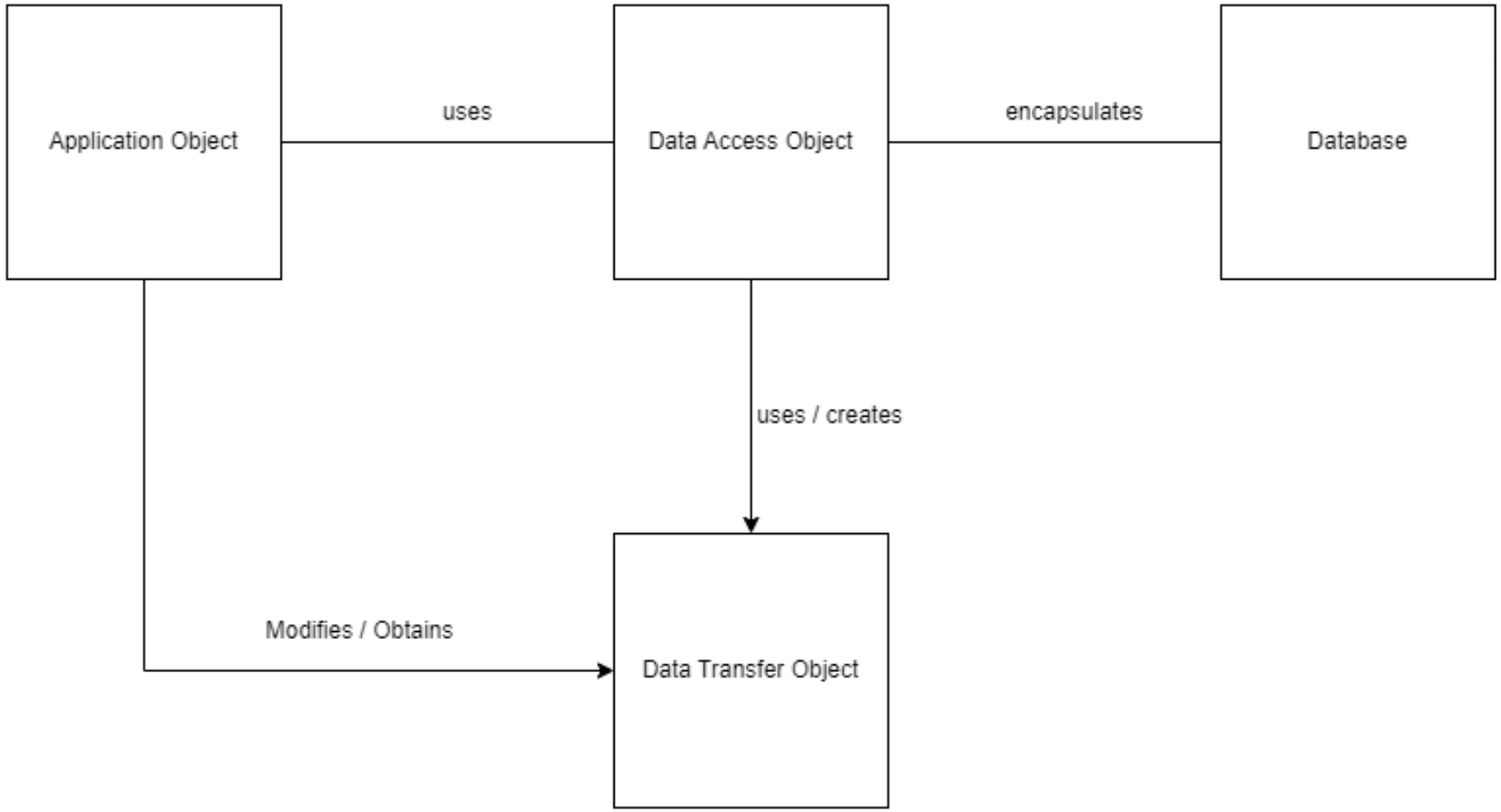
Data Access Object (DAO) – 1/2

- «Client» classes:
  - Application code that needs to access the database
  - Ignorant of database details (connection, queries, schema, ...)
- «DAO» classes:
  - Encapsulate all database access code (mysql-connector-python)
  - The only ones that will ever contact the database
  - Ignorant of the goal of the Client

Data Access Object (DAO) – 2/2

- Low-level database classes, to handle the connection (`MySQLConnection`, Pooled connection,…)
  - Used by DAO (only!) but invisible to Client
- «Transfer Object» (TO) or «Data Transfer Object» (DTO) classes
  - Contain data sent from Client to Dao and/or returned by DAO to Client
  - Represent the data model, as seen by the application
  - May use `@dataclass`
  - Ignorant of DAO, ignorant of database, ignorant of Client
  - The DTO acts as a data store that moves the data from one layer to another
  - Should implement the `__eq__()` and `__hash__()` functions using the primary key
  - May implement `__str__()` and other dunder methods as needed

DAO Diagram


Application Object
uses
Data Access Object
encapsulates
Database
uses / creates
Modifies / Obtains
Data Transfer Object

DAO: application example

```
from dao import StudentDAO
from dto import StudentDTO

student_dao = StudentDAO()

student1= "Lary"
student2 = "Mike"
school = "School:Hindenburg"

stud1 = StudentDTO(school,student1)
stud2 = StudentDTO(school,student2)

student_dao.add_student(stud1)
student_dao.add_student(stud2)

student_names = student_dao.get_all_student("School:Hindenburg")
print(student_names)
```

For example a @dataclass

The DAO then implements the methods to interface with the student table in the database

DAO design criteria

- DAO has no state
  - No instance variables (except Connection - maybe)
- DAO manages one ‘kind’ of data
  - Uses a small number of DTO classes and interacts with a small number of DB tables
  - If you need more, create many DAO classes
- DAO offers CRUD methods
  - Create, Read, Update, Delete
- DAO may offer search methods
  - Returning collections of DTO

DAO: example

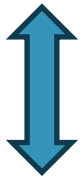
```python
class studentDAO:
    def __init__(self):
        #possibly a state to keep information about a connection
        # we will see this with pooling

    def get_methods(self):
        try:
            cnx = mysql.connector.connect(database='student')
        except mysql.connector.Error as err:
            print(err)
            result = None
        else:
            result = []
            cursor = cnx.cursor(dictionary=True)
            cursor.execute("SELECT * FROM students")
            for row in cursor:
                result.append(studentDTO(row["id"], row["name"]))
            cursor.close()
        finally:
            cnx.close()
            return result
```

OBJECT-RELATIONAL MAPPING

Object Relational Mapping (ORM)

- ***Object Relational Mapping*** is programming pattern that enables for moving data between objects and a database while keeping them independent of each other.
  - In the database, entities are represented as rows of a table, and they can be related to entries of other tables

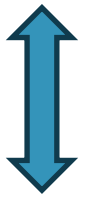


  - In the Python application, we represent entities as objects, and we need to represent their relationships

Object Relational Mapping (ORM)

Database,
table Pets

name	species	age	weight_in_kg	favorite_food
Pocket	dog	3	22.5	Kibbles & Bits
Mittens	cat	7	8.0	Fancy Feast: Salmon Edition
Mrs. Birdy III	bird	22	0.5	Froot Loops



DTO Pet

Class Pet

Mapping Tables to Objects

- Goal: guidelines for creating a set of (data)classes to represent information stored in a relational database: will be used as DTO
- Goal: guidelines for designing the set of methods for DAO objects

Tables → data class ORM rules

1. Create one dataclass per each main database entity
   - Except tables used to store n:m relationships!
2. Class names should match table names
   - In the singular form (Utente; User)
3. The Class should have one attribute for each column in the table, with matching names
   - According to Python naming conventions (NUMERO_DATI -> numero_dati)
   - Match the data type
   - Except columns used as foreign keys

Tables → data class ORM rules

4. Add the getter (`@property`) and setter (`@attr.setter`) methods for the attributes, if needed. The setter method cannot be specified if the dataclass uses the `frozen=True` parameter.
5. Define `__eq__()` and `__hash__()` using the **exact** set of fields that compose the primary key of the table

Relationships, Foreign keys → Class

- Define additional attributes in the DTO classes, for every relationship that we want to easily navigate in our application
  - Not necessarily *all* relationships!

Cardinality-1 relationship

- A relationship with cardinality **1** maps to an attribute referring to the corresponding Python object
  - not the PK value
- Use singular nouns.

1:1 relationship

```
STUDENTE
--------
matricola (PK)
fk_persona

@dataclass
class Studente:
      persona: Persona
      codice_fiscale: str
```

```
PERSONA
-------
codice_fiscale (PK)
fk_studente

@dataclass
class Persone:
      studente: Studente
      matricola: int
```

Cardinality-N relationship

- A relationship with cardinality **N** maps to an attribute containing a collection
  - The elements of the collection (for example list or set) are corresponding Python objects (not PK values).
  - Use plural nouns.
- The class should have methods for reading (get, …) and modifying (add, …) to the collection

1:N relationship

```
STUDENTE
------------------
matricola (PK)
fk_citta_residenza

@dataclass
class Studente:
      cittaResidenza: Citta
```

```
CITTA
---------------
cod_citta (PK)
nome_citta

@dataclass
class Citta:
      studentiResidenti: list[Studente]
```

1:N relationship

```
STUDENTE
------------------
matricola (PK)
fk_citta_residenza
```

```
CITTA
---------------
cod_citta (PK)
nome_citta
```

In SQL, there is no «explicit» Citta->Studente foreign key. The same FK is used to navigate the relationship in both directions.

```
@dataclass
class Studente:
        cittaResidenza: Citta
```

```
@dataclass
class Citta:
        studentiResidenti: list[Studente]
```

In Python, both directions (if needed) must be represented explicitly.

N:M relationship

```
ARTICLE
---------------
id_article (PK)
Article data…
```

```
AUTHORSHIP
---------------
id_article (FK,PK*)
id_creator (FK,PK*)
id_authorship (PK#)
```

```
CREATOR
---------------
id_creator (PK)
Creator data…
```

```
@dataclass
class Article:
    creators: set[Creator]
```

```
@dataclass
class Creator:
    articles: set[Article]
```

N:M relationship

In SQL, there is an extra table just for the N:M relationship .

```
ARTICLE
---------------
id_article (PK)
Article data…
```

```
AUTHORSHIP
---------------
id_article (FK,PK*)
id_creator (FK,PK*)
id_authorship (PK#)
```

```
CREATOR
---------------
id_creator (PK)
Creator data…
```

The PK may be an extra field (#) or a combination of the FKs (*)

```
@dataclass
class Article:
    creators: set[Creator]
```

```
@dataclass
class Creator:
    articles: set[Article]
```

The extra table is not represented.
The PK is not used.

Storing Keys vs Objects

`id_citta_residenza: int`

- Store the *value* of the foreign key
- Easy to retrieve
- Must call a read method from the DAO to get all the data
- Tends to perform more queries

`citta_residenza: Citta`

- Store a *fully initialized object*, corresponding to the matching foreign row
- Harder to retrieve (must use a Join or multiple/nested queries)
- Gets all data at the same time (eager loading)
- All data is readily available
- Maybe such data will not be needed

Storing Keys vs Objects (3rd way)

```
citta_residenza : Citta = field(default_factory=lambda: []) // lazy
citta_residenza : Citta = None                              // lazy
```

- Store a *partially initialized object*, with only the 'id' field set
  - Or even a null field
- Easy to retrieve
- Must ask the DAO to have the real data (lazy loading), but only once
- Loading details may be hidden behind getters

Identity problem

- It may happen that a single object gets retrieved many times, in different queries
  - Especially in the case of N:M relationships

```
articles = dao.list_articles()
for article in articles:
      authors= dao.get_creators_for(article)
      article.creators(authors)
```

```
…
authors = []
for row in cursor:
      authors.append(Creator(…))
…
return authors
```

Identity problem

- It may happen that a single object gets retrieved many times, in different queries
  - Especially in the case of N:M relationships

```
articles = dao.list_articles()
for article in articles:
        authors= dao.get_creators_for(article)
        article.creators(authors)
```

```
…
authors = []
for row in cursor:
        authors.append(Creator(…))

        authors
```

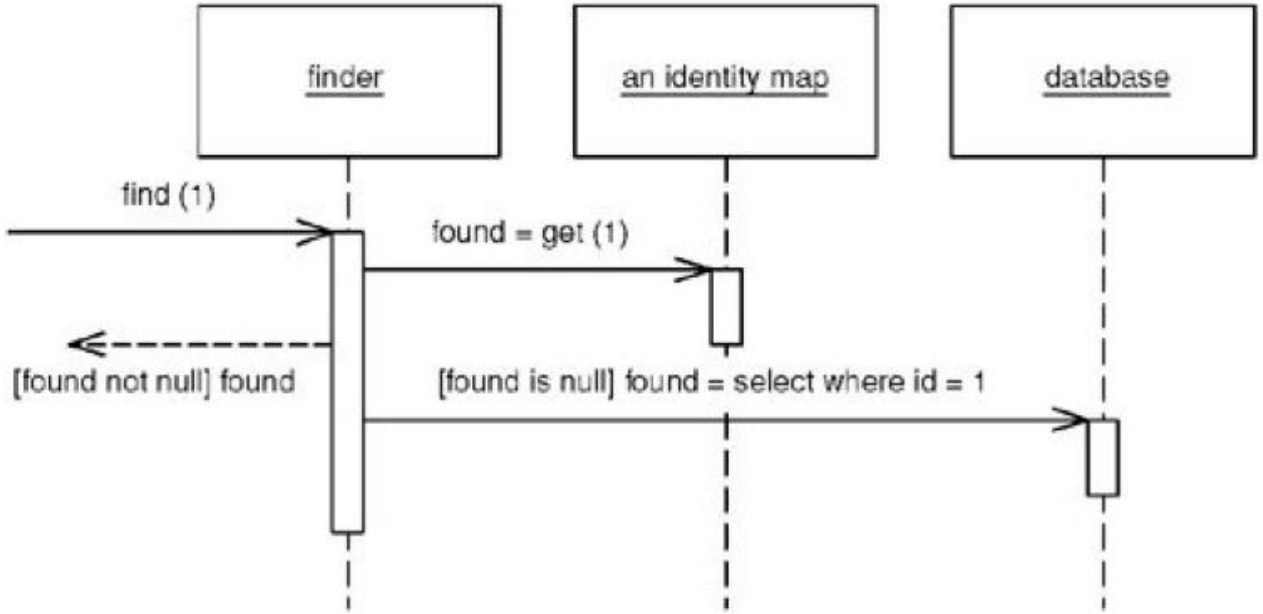
If the same Creator is author of many articles, we would repeatedly query the database to create new objects with identical information

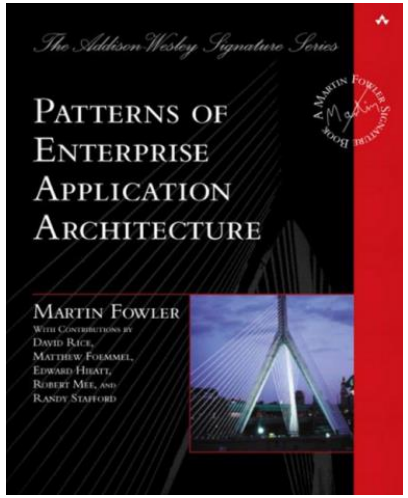
Identity problem

- It may happen that a single object gets retrieved many times, in different queries
  - Especially in the case of N:M relationships
- Different «identical» objects will be created
  - They can be used interchangeably
  - They waste memory space
  - They can’t be compared for identity (== or !=)
  - You can’t store additional information in those objects
- Solution: avoid creating pseudo-identical objects
  - Store all retrieved objects in a map (for example, using a dictionary)
  - Don’t create an object if it’s already in the map

Identity Map pattern

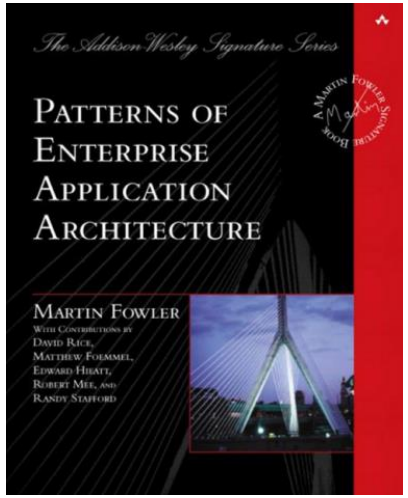
- Ensures that each object gets loaded only once, by keeping every loaded object in a map
- Looks up objects using the map when referring to them.





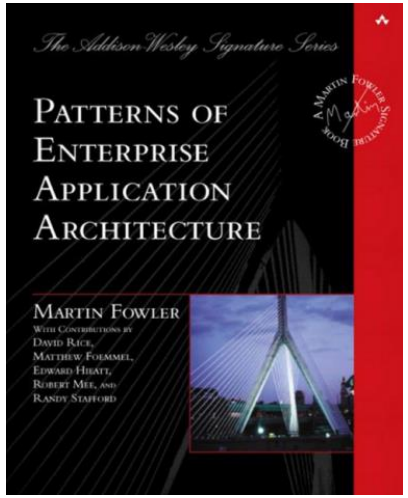
Creating an Identity Map

- One identity_map per database table
- The identity_map stores a dictionary
  - Key = field(s) of the Table that constitute the Primary Key
  - Value = Object representing the table



Using the Identity Map

- Create and store the identity_map in the Model
- Pass a reference to the identity_map to the DAO methods
- In the DAO, when loading an object from the database, first check the map
  - If there is a corresponding object, return it (and don't create a new one)
  - If there is no corresponding object, create a new object and put it into the map, for future reference
- If possible, check the map *before* doing the query



ORM Libraries

- There are many **Object Relational Mappers** in Python, that are libraries that implement the ORM logic and usually much more (they integrate the connector, implement DAO)

SQLAlchemy

django

peewee

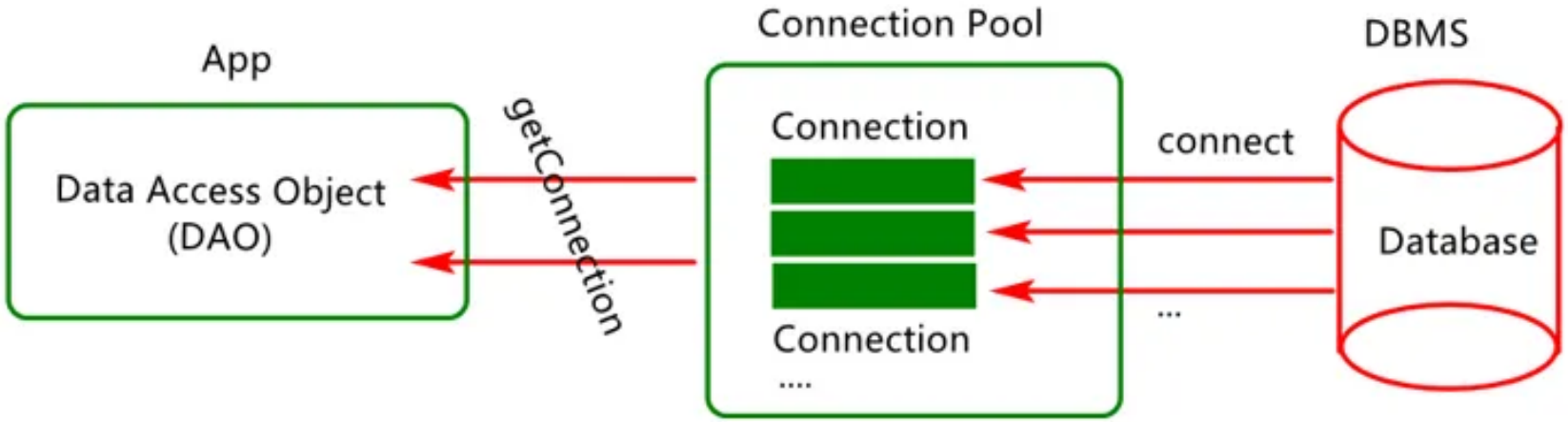
PONY Object-Relational Mapper

CONNECTION POOLING

Connection pooling

- Opening and closing DB connection is expensive
  - Requires setting up TCP/IP connection, checking authorization, …
  - After just 1-2 queries, the connection is dropped and all partial results are lost in the DBMS
- Connection pool
  - A set of “already open” database connections
  - DAO methods “lend” a connection for a short period, running queries
  - The connection is then returned to the pool (not closed!) and is ready for the next DAO needing it

Connection pooling conceptual model



Connection pooling with mysql-connector

- The `mysql.connector.pooling` module implements pooling.
- A pool opens a number of connections and handles thread safety when providing connections to requesters.
- A connection pool has several properties:
  - size : indicates the number of connections available in the pool. It is configurable at pool creation time and cannot be resized thereafter.
  - name: can be retrieved from the connection pool or connections obtained from it.
- It is possible to have multiple connection pools. This enables applications to support pools of connections to different MySQL servers, for example.
- For each connection request, the pool provides the next available connection. No round-robin or other scheduling algorithm is used. If a pool is exhausted, a `PoolError` is raised.

Creating a pool

```
cnxpool = mysql.connector.pooling.MySQLConnectionPool(pool_name = "mypool",
                                                      pool_size = 3,
                                                      database = "test",
                                                      user = "John",
                                                      password = "Wick",
                                                      host = "127.0.0.1")
```

The `cnxpool` object is an instantiation of the class `PooledMySQLConnection` . Differently from `MySQLConnection` objects, `PooledMySQLConnection` objects cannot be used directly as connections, but we must lend a connection from them

Lending a connection

- We can ask a connection from the pool using the `get_connection()` method
  - Warning: if there are no connections available this method raises a `PoolError` exception

```
cnxpool = mysql.mysql.connector.pooling.MySQLConnectionPool(pool_name = "mypool",
                                                            pool_size = 3,
                                                            database = "test",
                                                            user = "John",
                                                            password = "Wick",
                                                            host = "127.0.0.1")

cnx = cnxpool.get_connection()

//do operations

cnx.close()
```

`cnx` is an instantiation of the class `PooledMySQLConnection` . It is similar to a `MySQLConnection` object, but with one notable difference:

- The `close()` method return the connection to the pool, does not terminate it!

Benchmarks

# Iterations	1	10	100	1000	10000
Non-Pooling	0.012s	0.081s	0.717s	8.81s	92.2s
Pooling	0.031s	0.039s	0.081s	0.351s	2.42s

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